

Database Name: RUET

```
CREATE DATABASE IF NOT EXISTS Ruet;  
USE Ruet;
```

Step 1: Create the following tables with appropriate constraints:

-- Table: students

```
CREATE TABLE students (  
    student_id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    department VARCHAR(50) NOT NULL,  
    email VARCHAR(100) UNIQUE NOT NULL,  
    admission_year YEAR NOT NULL  
);
```

-- Table: faculty_members

```
CREATE TABLE faculty_members (  
    faculty_id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    department VARCHAR(50) NOT NULL,  
    designation VARCHAR(50),  
    email VARCHAR(100) UNIQUE NOT NULL,  
    hire_date DATE NOT NULL  
);
```

-- Table: office_workers

```
CREATE TABLE office_workers (  
    worker_id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100) NOT NULL,  
    position VARCHAR(50) NOT NULL,  
    department VARCHAR(50),  
    email VARCHAR(100) UNIQUE,  
    join_date DATE NOT NULL  
);
```

Step 2: Insert the following sample data

-- Insert into students

```
INSERT INTO students (name, department, email, admission_year) VALUES  
( 'Sumaiya Samawat', 'CSE', 'sumaiya@ruet.ac.bd', 2021),  
( 'Rakib Hossain', 'EEE', 'rakib@ruet.ac.bd', 2020),
```

('Nusrat Jahan', 'ME', 'nusrat@ruet.ac.bd', 2022);

-- Insert into faculty_members

```
INSERT INTO faculty_members (name, department, designation, email, hire_date) VALUES
('Dr. Mahmud Hasan', 'CSE', 'Professor', 'mahmud@ruet.ac.bd', '2010-03-15'),
('Dr. Farzana Akter', 'EEE', 'Associate Professor', 'farzana@ruet.ac.bd', '2015-06-01'),
('Dr. Anik Rahman', 'ME', 'Lecturer', 'anik@ruet.ac.bd', '2020-01-10');
```

-- Insert into office_workers

```
INSERT INTO office_workers (name, position, department, email, join_date) VALUES
('Kamal Uddin', 'Admin Officer', 'Administration', 'kamal@ruet.ac.bd', '2012-11-01'),
('Salma Begum', 'Accountant', 'Finance', 'salma@ruet.ac.bd', '2018-04-20'),
('Habib Rahman', 'Clerk', 'Academic', 'habib@ruet.ac.bd', '2021-07-05');
```

Step 3: Write queries to solve the following:

1. Show all students:

```
SELECT * FROM students;
```

2. List all faculty members from the CSE department:

```
SELECT * FROM faculty_members
WHERE department = 'CSE';
```

3. Show office workers who joined after 2015:

```
SELECT * FROM office_workers
WHERE join_date > '2015-01-01';
```

4. Count students by department:

```
SELECT department, COUNT(*) AS total_students
FROM students
GROUP BY department;
```

5. Find all faculty who are lecturers:

```
SELECT * FROM faculty_members
WHERE designation = 'Lecturer';
```

6. List students admitted after 2020:

```
SELECT * FROM students  
WHERE admission_year > 2020;
```

7. Count office workers per department:

```
SELECT department, COUNT(*) AS total_workers  
FROM office_workers  
GROUP BY department;
```

8. Find faculty members hired before 2015:

```
SELECT * FROM faculty_members  
WHERE hire_date < '2015-01-01';
```

9. Show student name and department only:

```
SELECT name, department FROM students;
```

10. List unique departments across all tables:

```
SELECT department FROM students  
UNION  
SELECT department FROM faculty_members  
UNION  
SELECT department FROM office_workers;
```